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AIML-B1

Experiment-7

K-Means Clustering for Customer Segmentation

Synopsis of Dataset

Dataset Description

The Mall Customers dataset is a widely used benchmark dataset in unsupervised machine learning, specifically for demonstrating customer segmentation using clustering algorithms. It simulates real-world mall customer data collected through membership cards issued at a supermarket mall. The dataset is designed to help businesses understand customer behavior and group them into meaningful segments for targeted marketing strategies.

The dataset contains records of 200 unique customers, each described by five attributes. It is clean, structured, and contains no missing values, making it ideal for direct application of clustering algorithms after minimal preprocessing.

Number of Samples and Features

Property	Value
Total Samples (Rows)	200
Total Features (Columns)	5
Numeric Features	4 (CustomerID, Age, Annual Income, Spending Score)
Categorical Features	1 (Gender)
Missing Values	None (0)
Female Customers	112 (56%)
Male Customers	88 (44%)

Description of Each Attribute

Attribute	Data Type	Range / Values	Mean	Description
CustomerID	Integer	1 – 200	100.5	Unique identifier for each customer
Gender	Categorical	Male / Female	–	Biological sex of the customer
Age	Integer	18 – 70 years	38.85 years	Age of the customer in years
Annual Income (k$)	Integer	$15k – $137k	$60.56k	Annual income of the customer in thousands of USD
Spending Score (1-100)	Integer	1 – 99	50.20	Score (1–100) assigned based on customer purchasing behavior and loyalty

Note: CustomerID is a non-informative identifier column and is excluded from clustering. Gender is a categorical feature and is also excluded for the numerical clustering analysis. The features used for clustering are Annual Income, Spending Score, and Age.